Henry Boyes
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SUMMARY

Data and ML engineer with production experience designing and deploying end-to-end machine learning pipelines for large-scale scientific data. Built and orchestrated a modular spatial proteomics pipeline processing ~3M cells across GPU/CPU HPC infrastructure (Slurm, A100), spanning deep learning inference, unsupervised phenotyping, and automated ETL workflows. Experienced in Python-based pipeline architecture, SQLite-backed job tracking, and structured data systems. Currently completing an M.S. in Data Science. Brings a foundation in process discipline and operational rigor from U.S. Navy quality management — applied directly to building reproducible, maintainable ML systems.

TECHNICAL SKILLS

Languages & ML: Python (scikit-learn, statsmodels, pandas, NumPy), R, SQL
ML & Modeling: Supervised/unsupervised learning, deep learning inference, clustering (K-means, hierarchical, Leiden/UMAP), dimensionality reduction (PCA), spatial statistics, feature engineering
Pipeline, Infrastructure, & Development: Git, Modular pipeline architecture, Slurm (HPC job orchestration), ETL workflows, GPU compute (A100), SQLite, CustomTkinter, openpyxl
Visualization & Reporting: matplotlib, seaborn, Tableau
Process Improvement: Lean Six Sigma Yellow Belt, project workflow optimization

PROFESSIONAL EXPERIENCE

Research Technician II – Cleveland Clinic Foundation

(Mar 2025 – Present)

- Designed and deployed a modular, production-grade spatial proteomics ML pipeline processing ~3M cells across multi-channel imaging datasets; stages span ingestion, preprocessing, GPU-accelerated deep learning segmentation (Cellpose v4), feature extraction, unsupervised phenotyping (Scanpy: Leiden clustering, UMAP), and spatial analysis (Squidpy). (github.com/boyeshenry-byte/akoya-pcf-pipeline)
- Engineered Slurm orchestration scripts to distribute workloads across GPU (A100) and CPU nodes on HPC infrastructure, enabling fully reproducible pipeline runs with minimal manual intervention.
- Built automated Python ETL workflows (pandas, NumPy) reducing data preparation time by 40%; standardized data flow from raw imaging inputs through structured AnnData/HDF5 outputs for downstream analysis.
- Implemented SQLite-backed pipeline status tracking and centralized logging to monitor job state across distributed compute stages, improving observability and enabling faster failure diagnosis.
- Managed a clinical research database of 800+ patient records with complex relational data structures; ensured data quality, integrity and compliance while supporting cross-functional oncology and clinical teams.
- Identified a legacy workflow bottleneck caused by a 47-sheet shared Excel log; designed and built a full-stack desktop application (Python, CustomTkinter, SQLite) with multi-user concurrent access, normalized schema, CRUD operations, filtered Excel export, and soft delete. (github.com/boyeshenry-byte/lab-accession-app)
- Conducted exploratory analysis on a high-dimensional clinical dataset (n=25) of bladder cancer patients receiving ICI therapy; applied logistic regression (scikit-learn) and ordinal regression (statsmodels) with bootstrapped confidence intervals to identify directionally significant biomarker associations with treatment response under small-sample constraints.

Research Assistant – Kent State University

(Aug 2023 – Oct 2024)

- Contributed statistical analysis to a peer-reviewed publication in *Acta Biomaterialia* (Aug 2025); designed analytical methodologies that improved data accuracy and processing consistency across the research team.
- Translated complex quantitative findings into accessible presentations for conference audiences and cross-disciplinary collaborators.

Hospital Corpsman 2nd Class, Quality Management – U.S. Navy, Naval Medical Center San Diego
(Dec 2018 – Apr 2020)

- Led data modernization project, transitioning from outdated legacy tracking systems to a new digital platform; improved HEDIS metrics by 60% through data-driven process redesign and performance monitoring.
- Analyzed performance data and designed optimization strategies that increased training readiness metrics from 60% → 95% within one month, demonstrating ability to drive measurable business outcomes.
- Supervised 2 junior staff; mentored and trained them, resulting in promotions.
- Recognized with Navy Achievement Medal and multiple performance awards.

Hospital Corpsman 3rd Class – U.S. Navy, Naval Medical Center San Diego
(Aug 2016 – Dec 2018)

- Maintained data accuracy and trained staff on documentation procedures, ensuring compliance with organizational standards.

Boatswain's Mate 3rd Class – U.S. Navy, USS Porter (DDG 78)
(Dec 2012 – Aug 2016)

- Managed project data and scheduling for 25+ personnel, tracking task completion and ensuring on-time delivery of mission-critical operations.

EDUCATION & CERTIFICATIONS

University of Pittsburgh (Expected 2027)
Master of Science – Data Science
Relevant Coursework - Machine Learning, Statistical Analysis, Database Systems
Kent State University (May 2024)
Bachelor of Science – Organismal Biology
Cum laude, GPA: 3.6/4.0
Associates of Science - Biology
GPA, 3.4/4.0
Lean Six Sigma Yellow Belt (2019) | Project Management Performance Excellence (2019)

AWARDS

Navy Achievement Medal (×4), Navy Good Conduct Medal (×2) | Departmental Award for Significant Academic Achievement — Kent State University, 2024